

Hardware Troubleshooting & Capstone Introduction

1. Download and run CPU-Z on your personal or lab PC, then take a screenshot of the CPU and Memory tabs and write down the processor model, RAM size, and RAM type detected by the tool.

->CPU-Z Hardware Information Task

1. Download and install **CPU-Z** from the official CPUID website.
2. Open CPU-Z after installation.
3. Select the **CPU** tab and record:
 - Processor name/model
 - Core speed
 - Number of cores and threads
4. Select the **Memory** tab and record:
 - RAM type (DDR4, DDR5, etc.)
 - RAM size
 - DRAM frequency
 - Channel mode, if shown
5. Take screenshots of both the **CPU** and **Memory** tabs.
6. Save the screenshots with clear labels such as:
 - CPU-Z_CPU_Tab
 - CPU-Z_Memory_Tab

Documentation Table

Information	Detected Value
Processor Model	
RAM Size	

Information	Detected Value
RAM Type	_____
RAM Frequency	_____
CPU Cores/Threads	_____

Important: Use the values displayed by CPU-Z on your actual PC rather than entering example specifications.

2. Simulate a hardware boot failure by unplugging a component (RAM or hard drive) from a test PC, power it on, and note the BIOS/POST beep code or error message shown. Use an online POST code reference to identify the likely hardware issue and document your findings.

Hint: If your PC uses visual POST codes, note the pattern or code number displayed.

->Hardware Boot Failure Simulation

For this practical, **use a lab/test PC only and make all hardware changes while the PC is completely powered off and unplugged**. Do not remove components while the system is running.

1. Record the normal condition

- Start the test PC normally and confirm that it boots.
- Note the motherboard model because POST codes vary between manufacturers.

2. Create the fault

- Shut the PC down completely and disconnect power.
- With the instructor/lab supervisor present, temporarily remove **one RAM module** from its slot.
- Reconnect power and start the PC.

3. Observe POST

Record what happens:

- Beep pattern from the motherboard speaker
- DRAM/Memory diagnostic LED

- Q-Code/POST code, if available
- Screen error message, if displayed

4. Interpret the result

For example, ASUS documentation states that a **DRAM Q-LED staying on indicates a memory/no-memory problem**, while its buzzer guide identifies **one long + two short beeps** as a memory anomaly. ([ASUS Global](#))

MSI also documents systems where **three long beeps indicate that no memory module is detected**. ([MSI USA](#))

5. Restore the PC

- Power the system off and unplug it.
- Reinstall the RAM firmly.
- Confirm the retaining clips lock.
- Start the PC and verify that it boots normally.

Documentation Table

Item	Observation
Test component	RAM
Fault created	RAM module temporarily removed
Beep/LED/POST code	_____
Error message	_____
Likely problem	Memory/RAM not detected
Reference used	Manufacturer's motherboard POST/diagnostic guide
Result after reinstalling RAM	_____

Important: Don't assume that a particular beep count applies to every motherboard. POST codes and beep patterns are manufacturer/model-specific, so record the actual code first and then compare it with the motherboard manufacturer's documentation. ASUS, for example, uses different Q-Codes for memory, graphics, CPU, and boot-device problems. ([ASUS Global](#))

3. Use Memtest86 (either bootable USB or Windows version) to perform a memory diagnostic on any available PC. Record any errors found and explain what kind of real-world problems (e.g., random crashes, blue screens) these errors could cause.

->**MemTest86 Memory Diagnostic**

1. Prepare MemTest86

- Download MemTest86 from the official PassMark website.
- Create a bootable USB drive using the MemTest86 image/installer.

2. Run the test

- Insert the USB into the test PC.
- Restart the computer and boot from the USB drive.
- Start the MemTest86 memory test.
- Allow the test to complete its recommended test passes.

3. Record the results

Record the information shown on the final results screen:

Item	Result
RAM Size	
RAM Type	
Test Passes	
Errors Found	
Test Status	PASS / FAIL
Failed Address(es)	

4. Understand the errors

If MemTest86 reports memory errors, faulty or unstable RAM can contribute to:

- Random application crashes
- System freezes or unexpected restarts
- Windows **Blue Screen of Death (BSOD)** errors

- Corrupted data or files
- Installation or decompression errors
- Failure to boot reliably

5. If errors are found

- Record the exact error count and test number.
- Take a screenshot/photo of the MemTest86 results.
- A memory error does **not automatically prove that the RAM stick itself is defective**; motherboard slots, memory settings, or other hardware can also contribute.

Documentation Example

MemTest86 Result: The memory diagnostic was completed successfully. The test reported **0 errors**, so no memory errors were detected during this test. The PC operated normally during the diagnostic.

If your actual test reports errors, replace **“0 errors”** with the exact number shown by MemTest86 and document the corresponding test/error information.

4. Given this scenario: A PC randomly shuts down during gaming, but works fine on basic tasks. Using HWiNFO or GPU-Z, check and record the GPU temperature and fan speeds under load. Based on the data, suggest a possible root cause and a troubleshooting step.

->GPU Temperature & Fan-Speed Troubleshooting

For this practical, run a game or GPU stress workload for several minutes and monitor the GPU using **HWiNFO** or **GPU-Z**.

Parameter	Idle	Under Load
GPU Temperature	_____ °C	_____ °C
GPU Fan Speed	_____ RPM / _____ %	_____ RPM / _____ %
GPU Utilization	_____ %	_____ %

Analysis

If the GPU temperature rises unusually high under gaming load **while the fan speed is also high**, overheating is a possible cause of the shutdown. A PC that is stable during basic tasks but shuts down during demanding gaming may also have a **power-supply or GPU power-delivery problem**, so temperature alone doesn't prove the cause.

Possible root cause: GPU overheating or insufficient/unstable power delivery under high load.

Troubleshooting step: First clean dust from the GPU and case airflow path and verify that all GPU power connectors are firmly attached. Then repeat the load test while monitoring temperature. If temperatures remain excessive, investigate GPU cooling and case airflow; if temperatures are normal but the PC still abruptly powers off, test the PSU and GPU power connections.

Documentation

Take a screenshot of HWiNFO/GPU-Z while the game is running and record the **maximum GPU temperature, fan speed, and GPU utilization** shown. This provides evidence for your diagnosis.

5. Imagine you are setting up a new office for a 30-person digital marketing agency. Using online research, select hardware specifications for 2 file servers and 3 workstations (CPU, RAM, storage, GPU if needed), estimate the total budget, and justify your choices for each role.

Hint: Consider typical tasks like video editing, file sharing, and backup needs.

->For a **30-person digital marketing agency**, I would separate the infrastructure into two file servers—one primary file/asset server and one backup server—and three representative workstations: two general marketing machines and one video-editing machine. Prices below are **indicative India prices for September 2026**, so they should be treated as a planning estimate rather than a quotation. Current Indian price-comparison sources also note that seller prices and availability can change. ([PriceRig](#))

1. File Server 1 — Primary File & Collaboration Server

Component	Recommended specification	Estimated cost
CPU	Intel Xeon E-2436, 6 cores/12 threads	₹45,000
RAM	64 GB ECC DDR5	₹1,20,000
OS/Cache storage	1 TB NVMe SSD	₹12,000

Component	Recommended specification	Estimated cost
Data storage	4 × 8 TB NAS/enterprise HDD, RAID 10	₹80,000
Motherboard	Server board with ECC + 10GbE support	₹45,000
PSU/Case/Cooling	Server-grade 650–750 W PSU + chassis	₹40,000
Estimated total		₹3,42,000

The Xeon E-2436 is a server-oriented 6-core/12-thread processor supporting ECC memory and up to 128 GB, making it appropriate for a continuously operating file server. ([Intel](#))

Why: This server would hold campaign files, video assets, shared documents and project folders. RAID 10 provides redundancy while retaining good read/write performance.

2. File Server 2 — Backup & Disaster-Recovery Server

Component	Recommended specification	Estimated cost
CPU	Intel Xeon E-2436	₹45,000
RAM	64 GB ECC DDR5	₹1,20,000
OS/Cache storage	1 TB NVMe SSD	₹12,000
Data storage	4 × 8 TB HDD, RAID 10	₹80,000
Motherboard	ECC-capable server motherboard	₹45,000
PSU/Case/Cooling	Server-grade	₹40,000
Estimated total		₹3,42,000

Why: The second server should not simply duplicate the first server's role. It should receive scheduled backups/snapshots of the primary server, protecting the agency against accidental deletion, drive failure or other data-loss events.

A NAS platform such as Synology's DS923+ demonstrates the type of functionality useful for a small business: file sharing, synchronization, backup and expandable storage. ([Synology](#))

3. Workstation 1 — Digital Marketing / General Office

Component	Specification	Estimated cost
CPU	AMD Ryzen 7 7700, 8C/16T	₹32,000
RAM	32 GB DDR5	₹8,000
Storage	1 TB NVMe SSD	₹8,000
GPU	Integrated graphics	₹0
Motherboard/PSU/Case	B650 + 650 W PSU + case	₹25,000
Estimated total		₹73,000

Why: Suitable for browser-heavy work, Photoshop/light graphics, social-media management, office applications, analytics dashboards and multitasking. The Ryzen 7 7700 has 8 cores/16 threads and a 65 W TDP. ([AMD](#))

4. Workstation 2 — Digital Marketing / Graphics

Component	Specification	Estimated cost
CPU	AMD Ryzen 7 7700	₹32,000
RAM	32 GB DDR5	₹8,000
Storage	1 TB NVMe SSD	₹8,000
GPU	NVIDIA RTX 4060 8 GB	₹30,000
Motherboard/PSU/Case	B650 + 650 W PSU + case	₹25,000
Estimated total		₹1,03,000

Why: The dedicated GPU provides additional capability for Photoshop, Illustrator, motion graphics and occasional video work. NVIDIA specifies the RTX 4060 with **8 GB GDDR6** memory and 3,072 CUDA cores. ([NVIDIA](#))

5. Workstation 3 — Video Editing / Content Creation

Component	Specification	Estimated cost
CPU	AMD Ryzen 7 7700	₹32,000
RAM	64 GB DDR5	₹16,000
Storage	2 TB NVMe SSD	₹15,000
GPU	NVIDIA RTX 4060 8 GB	₹30,000
Motherboard/PSU/Case/Cooling	B650 + 750 W PSU + case	₹35,000
Estimated total		₹1,28,000

Why: This machine gets more RAM and faster/larger SSD storage because video editing involves large project files, previews and media caches. The RTX 4060's 8 GB GDDR6 VRAM and hardware video features make it appropriate for a mid-range editing workstation. ([NVIDIA](#))

Budget Summary

Equipment	Quantity	Approx. subtotal
Primary file server	1	₹3,42,000
Backup server	1	₹3,42,000
General marketing workstation	1	₹73,000
Graphics workstation	1	₹1,03,000
Video-editing workstation	1	₹1,28,000
Total	5 systems	₹9,88,000

So I would allocate **approximately ₹10 lakh** for these five systems, with another **10–15% contingency** for networking, UPS units, additional storage, Windows/software licensing, installation and price changes. That puts the practical infrastructure budget at roughly **₹11–11.5 lakh**.

Important: These are representative systems for the assignment, not a recommendation that every employee needs a dedicated workstation of this specification. For a real 30-person office, the remaining employees could use lower-cost office PCs or laptops while connecting to the shared servers.

[AMD Ryzen 7 7700 Desktop Processor](#)

₹24,639.12

[ASUS GeForce RTX 4060 8GB](#)

₹29

[WD Red 4TB NAS Hard Drive](#)

₹14,400

[Intel Xeon E-2436 Processor](#)

₹44,940